

# URBAN E-COMMERCE DATA ANALYSIS

Unlocking Business Insights using SQL

Presented by: Ankita Dikondwar (Manchalwar)

# PROJECT OVERVIEW

- ❑ **Objective:** To analyze e-commerce sales, customer behavior, and product performance using SQL.
- ❑ **Goal:** Transform raw database records into actionable business insights.
- ❑ **Impact:** Help the business make data-driven decisions for marketing and inventory management.



# DATASET & SCHEMA OVERVIEW

The database consists of multiple interconnected tables.

Key Tables Used:

## Customers

- Sign-up details
- Location

## Products

- Categories
- Pricing & Stock Inventory

## Orders

- Order details
- Transaction info

# KEY BUSINESS QUESTIONS

- ❑ Who are our most valuable (top-spending) customers?
- ❑ Which products and categories generate the highest revenue?
- ❑ What is the month-over-month sales trend?
- ❑ Which geographical locations have the most orders?



# TECHNICAL APPROACH & SQL FEATURES

- ❑ **Data Aggregation:** SUM(), COUNT(), AVG() for revenue and order metrics.
- ❑ **Data Combining:** INNER JOIN and LEFT JOIN to connect customers, orders, and products.
- ❑ **Advanced Filtering:** GROUP BY, HAVING, and CASE statements.
- ❑ **Window Functions:** RANK(), DENSE\_RANK() to find top N products/customers

# KEY INSIGHT - SALES & REVENUE TRENDS

- ❑ Identified the peak sales months and seasonal purchasing trends.
- ❑ Discovered that the top 20% of products contribute to the majority of total revenue.
- ❑ Analyzed the average order value (AOV) across different customer segments.



# KEY INSIGHT - CUSTOMER BEHAVIOR

- ❑ Identified repeat customers vs one-time buyers.
- ❑ Categorized customers into 'High', 'Medium', and 'Low' spending tiers using SQL CASE Statements .
- ❑ Found the most preferred payment methods among active users.

# BUSINESS RECOMMENDATIONS

- ❑ **Targeted Marketing:** Send personalized offers to 'High-Tier' and repeat customers.
- ❑ **Inventory Optimization:** Maintain high stock levels for top-ranking products to prevent stockouts.
- ❑ **Customer Retention:** Introduce loyalty programs for one-time buyers to encourage repeat purchases.



The background is a gradient of dark blue to purple. It features several faint, overlapping circular patterns. Some of these circles have degree markings (140, 150, 160, 170, 180, 190, 200, 210, 220, 230, 240, 250, 260) and arrows, suggesting a compass or a circular scale. The text "THANK YOU" is centered in a large, white, sans-serif font.

THANK YOU